Events of the test bench

Event 1 — Reset (t = 0 to t = 8ns)

rst = 0 holds the PRF in reset. All 128 prf values are set to 0 and all ready bits are set to 1. At t = 8ns rst goes high and normal operation begins.

Event 2 — Allocate p32 and p33 (posedge at t = 25ns)

alloc_en = 0b0011, alloc_tag_0 = 32, alloc_tag_1 = 33 are driven at t = 18ns. On the posedge at t = 25ns, ready[32] = 0 and ready[33] = 0. This simulates the rename stage handing these registers out to two instructions. Their stored values are not touched — only the ready bits go low.

Event 3 — CDB writes p32 = 100 (posedge at t = 45ns)

cdb_valid = 1, cdb_tag = 32, cdb_data = 100 are driven at t = 38ns. On posedge at t = 45ns, prf[32] = 100 and ready[32] = 1. Since the read is combinational, rd_data_0 immediately becomes 100.

Event 4 — CDB writes p33 = 200 (posedge at t = 65ns)

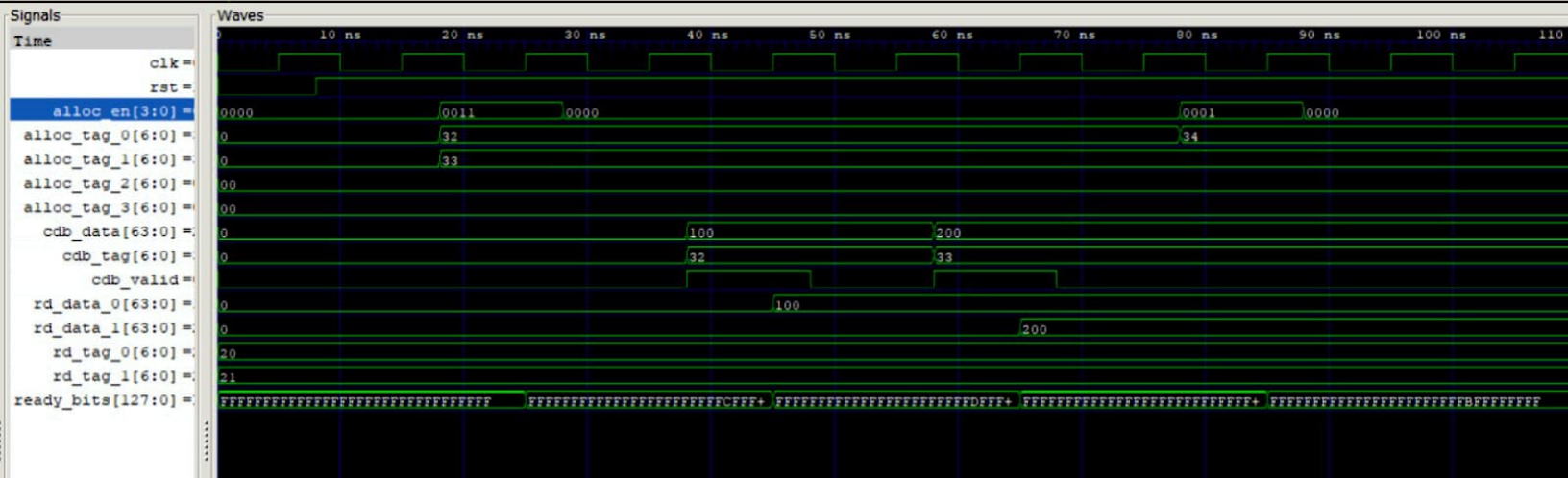
Same as Step 3 but for p33. On posedge at t = 65ns, prf[33] = 200 and ready[33] = 1. rd_data_1 immediately becomes 200.

Event 5 — Allocate p34 (posedge at t = 85ns)

alloc_tag_0 = 34, alloc_en = 0b0001 at t = 78ns. On posedge at t = 85ns, ready[34] = 0. This shows that allocating a new register does not disturb p32 and p33 — their ready bits and values remain unchanged.

Expected Signal Values

Time	Event	ready[32]	ready[33]	ready[34]	rd_data_0	rd_data_1
t = 0–8ns	Reset active	1	1	1	0	0
t = 25ns	Alloc p32, p33	0	0	1	0	0
t = 45ns	CDB writes p32=100	1	0	1	100	0
t = 65ns	CDB writes p33=200	1	1	1	100	200
t = 85ns	Alloc p34	1	1	0	100	200


Signals
Waves
Time
clk
rst
alloc_en[3:0]
alloc_tag_0[6:0]
alloc_tag_1[6:0]
alloc_tag_2[6:0]
alloc_tag_3[6:0]
cdb_data[63:0]
cdb_tag[6:0]
cdb_valid
rd_data_0[63:0]
rd_data_1[63:0]
rd_tag_0[6:0]
rd_tag_1[6:0]
ready_bits[127:0]
0
10 ns
20 ns
30 ns
40 ns
50 ns
60 ns
70 ns
80 ns
90 ns
100 ns
110
0000
0011
0000
0001
0000
0
32
34
0
33
00
00
0
100
200
0
32
33
0
100
0
200
20
21
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFFFFFFFFFFFFFFFCFFF+
FFFFFFFFFFFFFFFFFFFFFFFDFFF+
FFFFFFFFFFFFFFFFFFFFFFFFF+
FFFFFFFFFFFFFFFFFFFFFFFBFFFFFFFF